

Symmetric Root

- **Symmetric Properties of Roots**

Recall: if α and β are distinct roots of $ax^2 + bx + c = 0$, then $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$

- Certain relations involving α and β can also be determined from $\alpha + \beta$ and $\alpha\beta$, even when we do not know α and β distinctly. Such relations are said to be symmetric.
- They are symmetric in the sense that if α and β are interchanged, either the relation remains the same or is multiplied by -1

- Example 1
- If $\alpha \neq \beta$, determine whether or not each of the following is symmetric.
- (a) $\alpha + \beta$
- (b) $\alpha \beta$
- (c) $\alpha^2 + \beta^2$
- (d) $\alpha^2 - \beta^2$
- (e) $3\alpha + 2\beta$
- (f) $\alpha^3 + \beta^3$

- **Solution**

- (a) $\alpha + \beta = \beta + \alpha$
- $\alpha + \beta$ is symmetric
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- (b) $\alpha \beta = \beta \alpha$
- $\alpha \beta$ is symmetric
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- (c) $\alpha^2 + \beta^2 = \beta^2 + \alpha^2$
- $\alpha^2 + \beta^2$ is symmetric
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- (d) $\alpha^2 - \beta^2 = -(\beta^2 - \alpha^2)$
- $\alpha^2 - \beta^2$ is symmetric
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- (e) $3\alpha + 2\beta \neq 3\beta + 2\alpha$ since $\alpha \neq \beta$
- $3\alpha + 2\beta$ is not symmetric
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- (f) $\alpha^3 + \beta^3 = \beta^3 + \alpha^3$
- $\alpha^3 + \beta^3$ is symmetric

- Example 2

If α and β are the roots of the equation: $3x^2 - 4x - 1 = 0$, find the values of:

- (a) $\alpha + \beta$
- (b) $\alpha \beta$
- (c) $\alpha^2 + \beta^2$
- (d) $\frac{1}{\alpha} + \frac{1}{\beta}$
- (e) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
- (f) $\alpha^3 + \beta^3$
- (g) $\alpha - \beta$
- (h) $\frac{1}{\alpha + 1} + \frac{1}{\beta + 1}$

- **Solution**

- $3x^2 - 4x - 1 = 0$

- $a = 3, \quad b = -4, \quad c = -1$

- (a) $\alpha + \beta = \frac{-b}{a} = \frac{-(-4)}{3} = \underline{\underline{\frac{4}{3}}}$

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- (b) $\alpha \beta = \frac{c}{a} = \frac{-1}{3} =$

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- (c) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- Consider $(\alpha + \beta)^2$
- $(\alpha + \beta)(\alpha + \beta) = \alpha^2 + \alpha\beta + \alpha\beta + \beta^2$
- $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$
- $(\alpha + \beta)^2 - 2\alpha\beta = \alpha^2 + \beta^2$
- $\alpha^2 + \beta^2 = \frac{(4)^2}{9} - \frac{2(-1)}{3}$
- $\frac{16}{9} + \frac{2}{3} = \frac{16 + 6}{9} = \frac{22}{9}$

- (d) $\frac{1}{\alpha} + \frac{1}{\beta}$

- $\alpha \quad \beta$

- $\frac{\beta + \alpha}{\alpha \beta} \Rightarrow \frac{\alpha + \beta}{\alpha \beta} = \frac{4}{3} / \frac{-1}{3}$

- $\alpha \beta \quad \alpha \beta \quad 3 / 3$

- $\frac{4}{3} \times \frac{-3}{1} = -4$

- (e) $\alpha + \underline{\beta}$

- $\beta \quad \alpha$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \underline{(4/3)^2} - \underline{2(-1/3)}$$

- $\alpha \beta \quad \alpha \beta \quad - 1/3$

- $\underline{22} \div \underline{-1}$

- $9 \quad 3$

- $\underline{22} \times \underline{3} = \underline{-22}$

- $9 \quad 1 \quad 3$

- (f) $\alpha^3 + \beta^3 = (\alpha + \beta) (\alpha^2 + \beta^2 - \alpha \beta)$
- $= (\alpha + \beta) ((\alpha + \beta)^2 - 3\alpha\beta)$
- $= (4/3) (4/3)^2 - 3(-1/3)$
- $= (4/3) (16/9 + 1)$
- $= 4/3 (25/9) = 100/27$
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- (g) $\alpha - \beta$

- $\Rightarrow (\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta$

- $\alpha - \beta = \sqrt{(\alpha - \beta)^2 - 4\alpha\beta}$

- $= \sqrt{(4/3)^2 - 4(-1/3)}$

- $= \sqrt{\frac{16}{9} + \frac{4}{3}} = \sqrt{\frac{16 + 12}{9}}$

- $= \frac{\sqrt{28}}{3} = \frac{2\sqrt{7}}{3}$

- (h) $\frac{1}{\alpha + 1} + \frac{1}{\beta + 1}$
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- $\frac{\beta + 1}{(\alpha + 1)(\beta + 1)} + \frac{\alpha + 1}{(\alpha + 1)(\beta + 1)} = \frac{\alpha + \beta + 2}{\alpha\beta + \alpha + \beta + 1}$
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- $= \frac{4/3 + 2}{-1/3 + 4/3 + 1}$
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- $= \frac{4 + 6}{3} \div \frac{-1 + 4 + 3}{3}$
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- $= 10/3 \div 2$
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- $= 10/3 \times \frac{1}{2} = 5/3$
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- **FORMING QUADRATIC EQUATIONS WITH ALREADY GIVEN SUM AND PRODUCT**
- In forming a new quadratic equation from an existing sum and product, we first find the new sum and product and then substitute the values in the formula for quadratic equation which is
- **$x^2 - (\text{sum of roots})x + \text{product of roots} = 0$**
- **$x^2 - (\alpha + \beta)x + \alpha \beta = 0$**

- Example 5

If α and β are the roots of the equation:

$$3x^2 + 4x - 5 = 0,$$

form an equation in x whose roots are

- (a) α^2 and β^2
- (b) $1/\alpha$ and $1/\beta$
- (c) α/β and β/α
- (d) $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$
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- **Solution**

- $3x^2 + 4x - 5 = 0$

- $a = 3, b = 4, c = -5$

- If α and β are the roots, then

- Sum of roots $= \alpha + \beta = -b/a = -4/3$

- Product of roots $= \alpha \beta = c/a = -5/3$

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- (a) α^2 and β^2

- New sum $= \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha \beta$

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- $= (-4/3)^2 - 2(-5/3)$

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- $= \frac{16}{9} + \frac{10}{3} = \frac{16}{9} + \frac{30}{9} = \frac{46}{9}$

- $\frac{16}{9} + \frac{10}{3} = \frac{16}{9} + \frac{30}{9} = \frac{46}{9}$

- New product $= \alpha^2\beta^2 = (\alpha\beta)^2 = (-5/3)^2 = 25/9$
- New Quadratic Equation:
- $x^2 - (\text{sum of roots}) x + \text{product} = 0$
- $x^2 - 46/9x + 25/9 = 0$
- multiplying through by 9 gives
- **$9x^2 - 46x + 25 = 0$**

